

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA0713
NETWORKS

COURSE NAME: COMPUTER

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I Harish Kumar / 192521340 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Answer: Using the generator polynomial $X^3 + 1$ (bit pattern 1001) and message 11100011: append 3 zero bits (degree of the generator) to get 11100011000, then perform modulo-2 (XOR) polynomial long division by 1001. The division leaves a remainder of 100, so the transmitted message (data + CRC) is 11100011 100, i.e. 11100011100. (ii) If the leftmost bit is inverted, the received codeword becomes 01100011100. The receiver divides this by the same generator 1001; the division now leaves a non-zero remainder (010) instead of the expected all-zero remainder. Because a valid, error-free codeword always divides the generator polynomial exactly (remainder = 0), a non-zero remainder tells the receiver that an error occurred during transmission, so the frame is rejected/flagged for retransmission.

2. Answer: a) Throughput = (frames per second) $\times$ (bits per frame) = (12,000 frames/min $\div$ 60) $\times$ 10,000 bits = 200 frames/sec $\times$ 10,000 bits = 2,000,000 bps = 2 Mbps (this is the effective throughput even though the link's raw bandwidth is 10 Mbps, because only 12,000 of the theoretically possible frames get through each minute). b) Using Shannon's capacity formula, $C = B \times \log_2(1 + \text{SNR}) = 3000 \times \log_2(1 + 3162) = 3000 \times \log_2(3163) \approx 3000 \times 11.627 \approx 34,881$ bps ($\approx$ 34.88 kbps).

3. Answer: A Hamming (12,8) code takes 8 data bits and adds 4 parity bits (placed at bit positions 1, 2, 4 and 8, the powers of two) to form a 12-bit codeword, so it can detect and correct any single-bit error — exactly the kind of isolated bit-flip caused by occasional electromagnetic interference on the wireless link. For example, encoding the data bits 11010010 gives the 12-bit codeword 011010110010. If interference flips the bit at position 7 during transmission, the received word becomes 011010010010. At the receiver, three parity checks are recomputed; the pattern of which checks fail (the "syndrome") is 7 in binary, which directly points to bit position 7 as the corrupted bit. The receiver then simply flips that bit back, recovering the original heart-rate/blood-pressure data correctly and instantly — without needing to ask for a retransmission, which is exactly why an error-correcting code (rather than only error-detecting) suits this real-time medical-monitoring scenario.

4. Answer: a) With a /27 mask on 172.16.10.0/24, 3 extra bits are borrowed for subnetting (27 – 24 = 3), giving $2^3 = 8$ subnets, and 5 bits remain for hosts, giving $2^5 - 2 = 30$ usable hosts per subnet (block size = 32 addresses per subnet). The subnets are 172.16.10.0/27, .32/27, .64/27, .96/27, .128/27, .160/27, .192/27 and .224/27. The address 172.16.10.78 falls in the range 64–95, so it belongs to the 172.16.10.64/27 subnet, whose network address is 172.16.10.64 and broadcast address is 172.16.10.95. The address 172.16.10.95 also falls within this same subnet's range (64–95) — in fact it is exactly that subnet's broadcast address, so it lies inside the range but is not a usable host address. b) The address 2001:0db8:0000:0000:0000:ff00:0042:8329 compresses (dropping leading zeros in each group and replacing the longest run of all-zero groups with ::) to 2001:db8::ff00:42:8329, and expands back to its full form as 2001:0db8:0000:0000:0000:ff00:0042:8329. With a /64 prefix, the network prefix is the first 64 bits, i.e. 2001:0db8:0000:0000::/64 (2001:db8::/64 compressed), leaving 128 – 64 = 64 host bits, so the network can contain 2^{64} possible host addresses.

5. Answer: The primary networking problem is a flow-control mismatch (receiver overload) between the sender and the receivers: the exam server keeps transmitting at a high, steady rate that suits its own 10 Gbps link, without adapting to each student's much slower and widely varying access speed (100 Mbps fiber, 20 Mbps Wi-Fi, 5–10 Mbps mobile). Because the slower-connection devices cannot receive and process incoming packets as fast as the server

sends them, their receive buffers fill up and overflow; once a buffer overflows, further incoming segments are dropped, forcing the sender to retransmit them, which adds even more traffic to already-congested links and delays acknowledgments — producing exactly the incomplete downloads, repeated retransmissions and portal freezes described. This is the classic problem that flow control is designed to prevent: a mechanism such as TCP's sliding-window flow control, where each receiver advertises how much buffer space it currently has, lets the sender throttle its transmission rate to match the slowest/most constrained receiver's actual processing capacity, avoiding buffer overflow and the resulting retransmission storm.